



Latency, Availability & Rate Limits

Week 4 • Day 4 • Technical Architecture & Constraints

TPM Academy



Day 4 Objectives

- Distinguish **Service Level Indicator (SLI)** / **Service Level Objective (SLO)** / **Service Level Agreement (SLA)** / **error budget**
- Set the **three SLOs**: latency, availability, throughput / rate limit
- Walk the **latency budget** across your Container diagram — does the math fit?
- Design a **rate-limit policy** with explicit failure-mode behavior
- Name an **error-budget consequence** — the behavior change when we breach



Today's Shape

Clock	Block	Outcome
09:00 – 09:15	Opening	Pin Container diagram; recap SLI/SLO/SLA
09:15 – 10:30	Block 1 + Activity 1	SLO Triage
10:45 – 12:00	Block 2 + Activity 2	Set your three SLOs
13:00 – 14:15	Block 3 + Activity 3	Latency-budget walk
14:30 – 16:00	Block 4 + Activity 4 + Wrap	Cross-review + AI sanity check

Block 1 — The Vocabulary

SLI / SLO / SLA / Error budget



Get the Words Right

Term	Definition	Audience
SLI	Measurement: latency, availability, error rate	Operators / engineers
SLO	Target for the SLI	TPMs, eng leads
SLA	Contractual commitment with consequences	Customer-facing
Error budget	$(1 - \text{SLO})$ over the window	Eng leads, TPMs

Common error: "We commit to 99.99%" written casually = SLA, not SLO. Use SLO for internal targets; SLA only when contractually intended.

Three SLOs Are Usually Enough

Latency

" $p \leq \text{ms}$ % of the time over "

Anchor: when does the user give up?

Availability

"% of requests succeed over "

Anchor: cost of a minute of downtime?

Throughput / rate limit

"system handles rps per "

Anchor: what abuse / accident are we protecting against?

Each must have: a percentile, a window, and a defense.



SLO Failure Modes

Failure	Example
No window	"p95 < 400ms" (over what?)
Average instead of percentile	"avg < 500ms" (hides bimodal distributions)
No defense	"99.9%" (why this number?)
Aspirational	"99.99%" without multi-region active-active
User-mismatched	"p95 < 5s" for a typing-feedback feature

Activity 1 — SLO Triage

Quads • 45 minutes

8 SLO statements. Identify failures. Rewrite the 5 worst.

- 15 min: triage
- 15 min: rewrite (percentile + window + defense)
- 5 min: identify the most subtle failure



15 + 15 + 5

Block 2 — Set Your Three SLOs

Latency / Availability / Rate limit



Error Budget — the Decision Tool

If availability SLO is **99.9%**, error budget is **0.1%** — about **43 minutes** of downtime per 30 days.

- **Inside the budget** → ship features (more risk-tolerant)
- **Near or past the budget** → prioritize reliability (more risk-averse)

Converts SLO from abstract to load-bearing: a team that respects its error budget is doing SRE, even if not labeled as such.



Worked SLOs (FieldPulse reconcile)

```
### SLO – Latency
**Measurement:** Reconcile modal open time, mobile client.
**Target:** p95 ≤ 1000ms, 99% over 30 days.
**Defense:** Dispatchers tap reconcile 30+ times/shift; > 1s
crosses the "I'd rather use paper" threshold (Wk 1
interview Maria, Tier Sheet op-signal #2).
**Verification:** Synthetic transaction in Datadog.

### SLO – Availability
**Measurement:** Successful (non-5xx) reconcile submissions.
**Target:** 99.5% over 30 days.
**Error budget:** ~3.6 hrs/month.
**Defense:** Dispatch volume averages 200 reconciles/day; 30-min
outage during peak shift = ~50 affected dispatchers.
**Verification:** Pingdom + 5xx alerting.
**Budget consequence:** If 7-day rolling burns >50% of budget,
we freeze new feature merges and prioritize reliability.

### Rate-limit policy
**Per-user:** 10 reconcile-submits/sec, burst 30.
**Per-tenant:** 100 reconcile-submits/sec.
**Failure mode:** HTTP 429 with Retry-After header; client
implements exponential backoff with jitter.
**Defense:** Protects against runaway script bug seen in the
2024 dispatcher-tool incident.
**Verification:** Load test before launch; rate-limit cap dashboard.
```

Activity 2 — Set Your Three SLOs

Quads • 50 minutes

- 15 min: latency SLO (percentile + window + defense)
- 10 min: availability SLO + error-budget translation
- 15 min: rate-limit policy with failure-mode behavior



15 + 10 + 15

Block 3 — Latency Budget Walk

Does the math fit your Container diagram?



Per-Hop Latency Reference (rough)

Hop type	Typical range
Network round-trip (LAN → Internet)	20–80ms
Database read (indexed)	5–30ms
Database write (single-row)	10–50ms
External API call (sync)	50–500ms (varies)
Async publish (Event Hubs/Service Bus local)	1–10ms
JWT validation	1–5ms
TLS handshake (cold)	50–200ms

These are starting points — even a Transport Layer Security (TLS) handshake varies by platform. Use real platform numbers if available.



The Latency Math

SLO target: $p95 \leq 1000ms$

Path: Mobile → API GW → Reconcile → Postgres read
→ Tickets module
→ Postgres write
→ Audit publish
→ response back

Estimated:

Mobile-API GW round trip: 80ms

API GW → Reconcile: 20ms

Postgres read:

30ms

Tickets module call (sync): 100ms ← dominates

Postgres write:

40ms

Audit publish (async): 10ms

Reconcile → API GW response: 20ms

Response back:

80ms

~380ms median

p95 likely ~700-900ms ✓

If math doesn't fit: tighter SLO / fewer hops / faster hops / wider percentile.

Activity 3 — Latency-Budget Walk

Quads • 50 minutes

Annotate your Container diagram with per-hop latency. Sum the path. Decide what gives if it doesn't fit.

- 15 min: annotate per-hop latency
- 10 min: sum the path
- 10 min: if doesn't fit, decide what gives
- 5 min: flag question for engineering



15 + 10 + 10 + 5

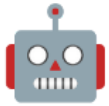
Block 4 — Cross-Review + AI Sanity Check

Three questions; one prompt; honest revision

The Three Cross-Review Questions

1. **Latency target realistic** given platform precedent?
2. **Availability target realistic** given the team's on-call reality?
3. **Rate-limit policy** protects against a real abuse / accident scenario?

Author quad listens, captures, decides. Same Week-3 review discipline.



AI Sanity Check

Role: Senior reliability engineer reviewing a feature's SLO sheet.

Context: <paste SLO sheet + latency walk + rate-limit policy>

Architecture stance: <paste TCD Section 1>

Task: Identify the top 3 reasons the SLOs as written might not be sustainable.

Constraints:

- Use only the information provided; do not invent platform details
- Be specific: which SLO, which scenario, which condition
- Suggest a more realistic alternative for each

Format: Numbered list – Issue / Scenario / Suggested target.

Activity 4 — Cross-Review + AI

Quad pairs • 55 minutes • Wrap

- 20 min: cross-review (3 questions)
- 15 min: AI sanity-check + adopt/defer/reject
- 10 min: revise Section 4 + provenance note



20 + 15 + 10 · 15-min wrap



Day 4 Wrap

What we built

- SLI / SLO / SLA / error-budget vocabulary
- Three SLOs (latency / availability / rate)
- Latency-budget walk through Container diagram
- Error-budget consequence stated
- Technical Concept Document (TCD) Section 4 drafted

What opens tomorrow

- **Technical trade-offs & constraints**
- Friday: assemble TCD Sections 5–6
- Stakeholder + sign-off matrix
- Cross-quad review of the full TCD

Bring to Day 5: Your TCD Sections 1–4. Tomorrow integrates everything into the final TCD.

Questions & Reflection

Which of your SLOs would you fold under pressure from a stakeholder asking for "just a bit more"?